Supplementary Material for: Temporal decoupling of CTLA-4 and ERK sensitivity in brain tumour signaling: a theoretical basis for sequencing checkpoint and MAPK inhibition

Table S4: Performance comparison of SVM and RF models across entities.

Entity (Name)	Model	MSE	R^2
Entity 1 (CTLA-4)	Random Forest	0.00094	0.92988
	SVM	0.01043	0.2189
Entity 2 (CD80)	Random Forest	0.00076	0.93777
	SVM	0.00596	0.51508
Entity 3 (PP2A)	Random Forest	0.00114	0.95482
	SVM	0.00541	0.78534
Entity 4 (RTKs)	Random Forest	0.00010	0.99048
	SVM	0.00352	0.67892
Entity 5 (PI3K)	Random Forest	0.00065	0.93695
	SVM	0.01166	-0.13932
Entity 6 (PTEN)	Random Forest	0.00276	0.91686
	SVM	0.00846	0.74537
Entity 7 (SMO)	Random Forest	0.00086	0.96181
	SVM	0.00433	0.80828
Entity 8 (PDK1)	Random Forest	0.00013	0.99243
	SVM	0.00511	0.71235
Entity 9 (AKT)	Random Forest	0.00139	0.96810
	SVM	0.00847	0.80496
Entity 10 (mTOR)	Random Forest	0.00310	0.92556
	SVM	0.00761	0.81696
Entity 11 (NF2)	Random Forest	0.00144	0.96031
	SVM	0.00879	0.75839
Entity 12 (Grb2)	Random Forest	0.00331	0.91969
	SVM	0.00869	0.78905
Entity 13 (SOS)	Random Forest	0.00274	0.94894
	SVM	0.00759	0.85830
Entity 14 (Ras)	Random Forest	0.02940	0.43163
	SVM	0.03601	0.30376
Entity 15 (Raf)	Random Forest	0.00325	0.89507
	SVM	0.00619	0.79989
Entity 16 (MEK)	Random Forest	0.01447	0.82471
	SVM	0.01559	0.81107
Entity 17 (ERK)	Random Forest	0.00069	0.98050

Entity (Name)	**Model**	**MSE**	**R^2**
	SVM	0.00427	0.87849
Entity 18 (p16)	Random Forest	0.00945	0.89738
	SVM	0.01349	0.85345
Entity 19 (CyclinD)	Random Forest	0.00023	0.92814
	SVM	0.00838	-1.60045
Entity 20 (CDK4)	Random Forest	0.00001	0.84530
	SVM	0.00531	-148.30962
Entity 21 (RB)	Random Forest	0.00278	0.96825
	SVM	0.00743	0.91509
Entity 22 (E2F)	Random Forest	0.00110	-0.44998
	SVM	0.00267	-2.52152
Entity 23 (G1-S)	Random Forest	0.00283	0.89871
	SVM	0.00602	0.78489
Entity 24 (CyclinE)	Random Forest	0.00174	-0.00500
	SVM	0.00201	-0.15939
Entity 25 (CDK2)	Random Forest	0.00029	0.98301
	SVM	0.00387	0.77684
Entity 26 (MDM2)	Random Forest	0.00076	-3.64810
	SVM	0.00329	-19.24900
Entity 27 (p53)	Random Forest	0.00084	0.95827
	SVM	0.00443	0.77976
Entity 28 (p21)	Random Forest	0.00061	0.96108
	SVM	0.00824	0.47308
Entity 29 (p27)	Random Forest	0.00253	0.95274
	SVM	0.01433	0.73257